ASEAN
GEOAI FUSION 2026

Module 7:
Advanced Knowledge: Advanced Geospatial Techniques in GeoAI Applications (AD1002)

AGAIF 2026 Boot Camp 2
Level: Advanced Knowledge of GeoAI

Thursday | 23 July 2026

Delivered by:
Ts. Naziffa Raha Md Nasir
Universiti Tenaga Nasional (UNITEN)

Session 7:

From GeoAI Insights to Impact

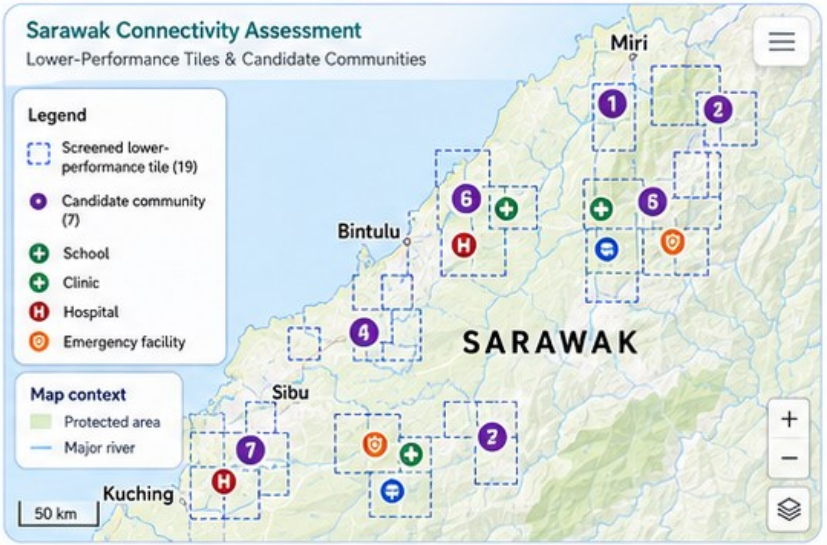


MVP Demonstration

Show the final interactive map and click-through evidence popups that turn the analysis into a decision-support GeoAI prototype for further assessment in Sarawak.

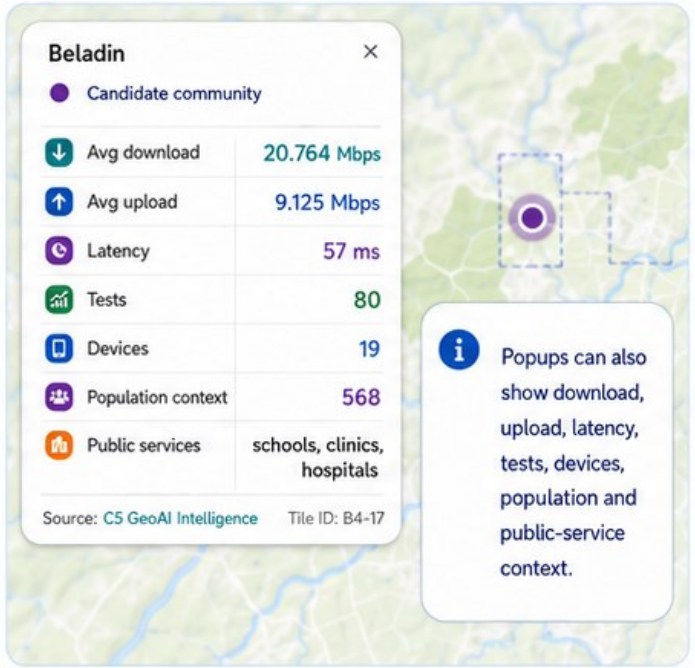
Canvas Reference:
C9 Value & Impact

1 Final interactive map



- Displays 19 screened performance tiles
- Highlights 7 candidate communities
- Supports quick spatial review across Sarawak

2 Click-through evidence popups



3 Why this prototype matters

- brings evidence into one inspectable interface
- helps planners review candidate communities quickly
- supports transparent follow-up assessment and validation
- transforms analysis outputs into a usable GeoAI MVP

This is a decision-support prototype for prioritisation and further assessment — not a final infrastructure deployment decision.

Key takeaway Session 7 begins with a working MVP: an interactive evidence map that makes the GeoAI workflow *inspectable, communicable* and ready for decision-focused discussion.

What Does the Evidence Actually Tell Us?

Interpreting results responsibly: what the data shows, and what it does not.

We combine performance, evidence strength and population context — and test how results change under different assumptions.

Canvas Reference:
C8 Responsible GeoAI
C9 Value & Impact

A Candidate Evidence Snapshot

We identified 7 candidate communities with stronger evidence for further assessment.

Rank	Community	Avg Download (Mbps)	Avg Upload (Mbps)	Latency (ms)	Tests	Devices	Est. Population
1	Kg. Meluan, Miri	12.45	5.21	58	28	9	29,640
2	Long Bawan, Marudi	13.02	5.80	61	23	7	18,920
3	Ulu Linau, Bintulu	14.10	6.03	59	21	6	15,310
4	Nanga Merit, Kapit	15.11	6.40	64	20	6	12,870
5	Belaga, Belaga	16.40	6.90	66	18	6	11,540
6	Long Tungan, Baram	17.20	7.05	67	16	5	9,830
7	Ba' Kelalan, Lawas	18.10	7.60	63	16	5	8,420

All seven communities are matched from OSM settlements and include population and public-service context for interpretation.

B Sensitivity / Robustness Check

We tested three evidence-strength scenarios to understand how results change.

Scenario	Rule (Tests / Devices)	Result (Communities)
Broader	≥10 tests and ≥3 devices	19 communities
Baseline	≥15 tests and ≥5 devices	7 communities
Stricter	≥20 tests and ≥7 devices	0 communities

Results vary under different assumptions. This is why evidence must be interpreted with transparency and caution.

C What We Can and Cannot Claim

What We CAN Say

- ✓ These locations show lower performance with stronger evidence.
- ✓ The data suggests these communities warrant further assessment.
- ✓ Evidence is derived from real user measurements and open datasets.

What We CANNOT Say

- ✗ These communities are confirmed underserved.
- ✗ There is no mobile coverage.
- ✗ A new tower or infrastructure is required.
- ✗ Exact infrastructure location or technical cause is identified.

Responsible GeoAI means clear boundaries between evidence, inference and decision.

Key takeaway

The evidence highlights 7 priority communities for further assessment.
Decisions must always be supported by additional validation and authoritative data.

From GeoAI Output to Value & Impact

Show how the GeoAI prototype moves from evidence outputs to practical planning value and potential real-world impact.

Canvas Reference:
C9 Value & Impact

1 GeoAI Outputs



- Shortlist of candidate communities
- Interactive HTML map
- Evidence table & popups

- screened areas for further assessment
- visual evidence for discussion
- decision-ready summary outputs

2 Outcome for Decision-Makers



- Focused validation
- Priority review discussion
- Better planning attention

- focuses attention on 7 candidate communities
- supports validation planning
- turns evidence into a usable planning conversation

3 Potential Impact



Operational
more focused field follow-up

Social
better support for affected communities

Strategic
stronger evidence for digital inclusion planning

Important claim boundary

This workflow identifies candidate communities for further assessment. It **does not** prove no coverage, infrastructure cause, or exact deployment need.

Key takeaway

The value of GeoAI is not only the map itself, but how the output helps stakeholders focus action, validate priorities, and frame evidence-based planning decisions.

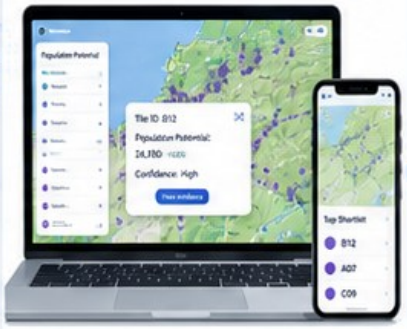
From Prototype to Operational GeoAI

Move from the MVP into a validated and deployable operational workflow

The prototype is the starting point. Real deployment requires pilot validation, official data integration, field checks and implementation planning.

Canvas Reference:
C10 Implementation Roadmap

1 MVP Prototype



- interactive HTML map
- evidence popups and shortlist
- ready for discussion and testing

2 Pilot Validation



- review with planners and regulators
- test workflow on real use cases
- confirm practical relevance

3 Official Data Integration



- operator and regulator datasets
- mobile coverage and infrastructure records
- strengthen evidence quality

4 Field & Technical Assessment



- field measurement and site checks
- engineering feasibility review
- assess capacity, backhaul and cost

5 Operational Deployment



- implementation plan and ownership
- update cycle and monitoring
- support real planning decisions

Key takeaway

A GeoAI MVP becomes operational only after validation, official data enrichment, technical assessment and implementation planning.

Canvas 1–10 Summary | Evidence-to-GeoAI Workflow for Connectivity Assessment in Sarawak

AGAIF Module 7 guided case | Theme: II — Industry Improvement

SESSION 5

DEFINE
(Canvas 1–4)

Define the problem, context and readiness foundation.

1 Canvas 1 — Problem Statement

- Real problem grounded using Scopus AI, original papers, ASEAN Digital Masterplan 2030, and ITU evidence
- **Decision question:** Which mapped communities show stronger evidence of comparatively lower observed mobile broadband performance and should be prioritised for further connectivity assessment?
- **Area of interest:** Sarawak, Malaysia
- **Theme:** Industry Improvement

2 Canvas 2 — Stakeholders & Users

- **Data providers:** Ookla, OpenStreetMap, WorldPop
- **Immediate users:** GeoAI analyst / participant team
- **Downstream users:** operator, regulator, planning stakeholders
- **Governance note:** actual operational decision owner not yet formally confirmed

3 Canvas 3 — Spatial Context

- **Area of interest:** Sarawak, Malaysia
- **Observation unit:** Ookla Q1 2026 mobile-performance tile
- **Decision unit:** mapped town / village / hamlet
- **Spatial match:** point-in-polygon intersection
- **Population context** aggregated by tile

4 Canvas 4 — Data Requirements

MUST-HAVE READY	Ookla Q1 2026 mobile data, OSM mapped settlements, download speed, tests, devices
SUPPORTING READY	upload speed, latency, WorldPop 2025, literature and policy evidence
GAPS / FUTURE VALIDATION	authoritative mobile coverage, operator infrastructure, field validation, official settlement registry, official community population, engineering feasibility, deployment cost

Ookla global records downloaded: **3,186,269** | Sarawak tiles analysed: **5,912** | OSM mapped settlements extracted: **1,708**

SESSION 6

DESIGN & ANALYSE
(Canvas 5–8)

Design the workflow, run the analysis and test robustness.

5 Canvas 5 — GeoAI Intelligence

- Converted kbps to Mbps
- Calculated Sarawak download quantiles
- Baseline lower-performance threshold: lowest 20% = **23.53 Mbps**
- Evidence-strength logic using tests and devices

6 Canvas 6 — GeoAI Solution Design

Ookla + OSM + WorldPop → filter Sarawak → classify lower performance → apply evidence thresholds → match communities → shortlist candidates

- Baseline rule: download in lowest 20%, tests ≥ 15, devices ≥ 5
- Matched communities using exact spatial intersection
- Added tile-level population context

7 Canvas 7 — Technology Stack

Scopus AI	Original research papers
ASEAN Digital Masterplan 2030	ITU evidence
Overpass Turbo	GeoLibre
Google Colab / Python	GeoPandas, Rasterio, Folium

GeoJSON / CSV / HTML outputs

8 Canvas 8 — Ethics, Privacy & Sustainability

Scenario	Threshold (Mbps)	Min Tests	Min Devices	Selected Tiles	Matched Communities
Broader	30.90	11	4	59	19
Baseline	23.53	15	5	19	7
Stricter	10.97	31	11	0	0

- Results support further assessment only
- Not authoritative coverage or infrastructure-deployment conclusions
- Public-service layer added as supporting context only
- Sensitivity, assumptions, data gaps, claim boundaries, and responsible interpretation must be made explicit

SESSION 7

DELIVER & COMMUNICATE
(Canvas 9-10 + integration)

Deliver outputs, explain insights and integrate for decision support.

9 Canvas 9 — Value & Impact

OUTPUT
- 19 screened lower-performance tiles
- 7 candidate mapped communities
- Interactive HTML assessment map
- Evidence table and export files

OUTCOME
Focused shortlist for validation planning and further assessment

POTENTIAL IMPACT
Better planning and evidence-based prioritisation after operator / regulator / field validation

10 Canvas 10 — Implementation Roadmap

1. Pilot validation with operator / regulator data
2. Field measurement and operational checking
3. Engineering feasibility and intervention assessment
4. Operational planning and phased deployment decision support
5. Documentation, provenance, and export package ready for next steps

INTEGRATION — Final Analytical Outputs

Seven Candidate Communities
1. Kampung Hulu
2. Kampung Seberang Maludam
3. Kampung Seberang Sebuyau
4. Kampung Tebelu
5. Maludam
6. Beladin
7. Kampung Simpok

Output Artifacts
- GeoJSON
- CSV
- HTML map

Workflow review
- GeoLibre visual review
- Colab notebook workflow

PUBLIC-SERVICE EXTENSION COMPLETED

Extracted mapped schools, clinics, hospitals, and emergency facilities from OpenStreetMap using Overpass Turbo → intersected facilities with screened tiles → attached tile-level service counts and names to candidate communities → did not use public services to change the original ranking

KEY TAKEAWAYS

1. The workflow began with evidence-based problem validation, not assumptions.
2. Open datasets were transformed into a traceable GeoAI screening pipeline.
3. Lower performance was interpreted together with observation strength.
4. The prototype narrowed thousands of observations to a defensible shortlist for further assessment.
5. Important operational gaps remain and must be validated before real deployment decisions.

ASEAN
GEOAI FUSION 2026

THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026